

Task 3: Hypothesis Testing

COURSE NAME

CSDS 413 Introduction to Data Analysis

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October 13, 2025

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Context

Car sales people usually claim that Japanese cars have better mileage than US cars. As a critical thinker, you know not to immediately believe what somebody who is trying to sell you something says. However, as a data scientist, you are curious to see whether this claim has any statistical basis.

You found some data collected in “cars.csv”. The first column contains a sample of measured mileage per hour for different US cars and the second column contains a sample of measured mileage per hour for different Japanese Cars. Answer the following questions:

- (a) How many US and how many Japanese cars does the dataset contain?
- (b) State the hypothesis you are testing to scrutinize the sales person's claim.
- (c) What is the statistic you would compute to test this hypothesis and its theoretical distribution?
- (d) What is the value of your statistic and what is its p-value?
- (e) What is your conclusion about the sales person's claim?

Note: Be sure to check any conditions or assumptions of the chosen test before computing statistics and drawing your conclusion.

1 Part A

How many US and how many Japanese cars does the dataset contain?

2 Part B

State the hypothesis you are testing to scrutinize the sales person's claim.

3 Part C

What is the statistic you would compute to test this hypothesis and its theoretical distribution?

4 Part D

What is the value of your statistic and what is its p-value?

5 Part E

What is your conclusion about the sales person's claim?
